

OSSP PRACTICAL 6

```
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical6
alurusaranya@LAPTOP-0PA61QEN:~$ cd practical6
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ mkfifo myfifo
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ls -l myfifo
prw-rw-r-- 1 alurusaranya alurusaranya 0 Sep 18 04:37 myfifo
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ nano server.c
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ nano client.c
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ gcc server.c -o server
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ls
client.c myfifo server server.c
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ cd practical6
./server
bash: cd: practical6: No such file or directory
./server
^C
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./server
^C
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ gcc client.c -o client
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ gcc client.c -o client
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ls
client client.c myfifo server server.c
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./server
^C
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ls
client client.c myfifo server server.c
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./server
```

```

root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ cd ~/practical6
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./client
Client:

```

```

alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ cd ~/practical6
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./client
Client: hello
Client: hi
Client: testing
Client: exit
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ |

```

```

alurusaranya@LAPTOP-0PA61QEN:~/practical6$ ./server
Server received: hello
Server received: hi
Server received: testing
Server received: exit
alurusaranya@LAPTOP-0PA61QEN:~/practical6$ |

```

A **Named Pipe (FIFO)** is an inter-process communication (IPC) mechanism used to exchange data between two processes. Unlike an ordinary pipe, a named pipe has a name in the file system and can be accessed by unrelated processes.

A FIFO can be created using the `mkfifo` command or the `mkfifo()` system call. One process can act as a **server** and read data from the FIFO, while another process acts as a **client** and writes data to it.

The server opens the FIFO in **read mode**, and the client opens it in **write mode**. Data written by the client is transferred through the FIFO and received by the server.

Working

1. Create a FIFO using `mkfifo`.
2. Compile the server program.

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3. Compile the client program.
4. Run the server in one terminal.
5. Run the client in another terminal.
6. The client sends messages through the FIFO.
7. The server reads and displays the messages.
8. The communication can be terminated by sending exit.